

# Customer Churn Analysis

*Business Report — Telecom / Subscription Customer Base*

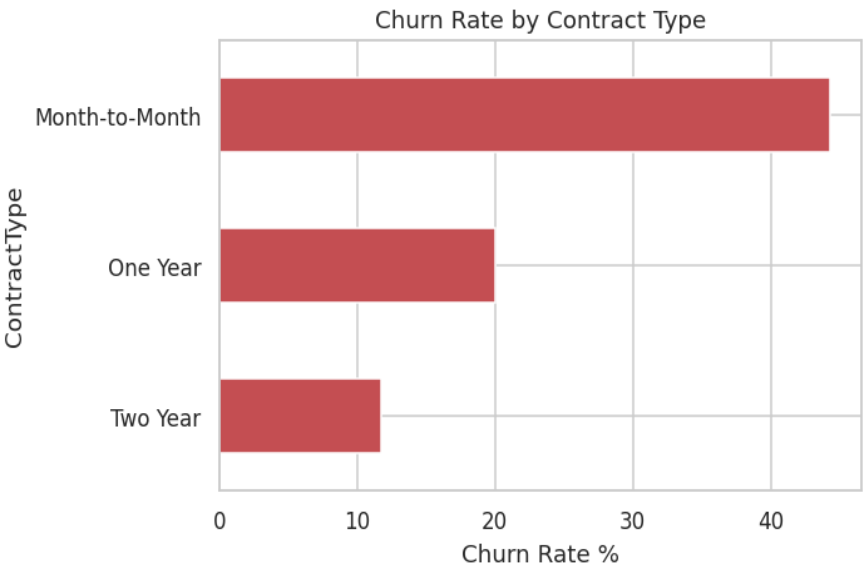
## Overview

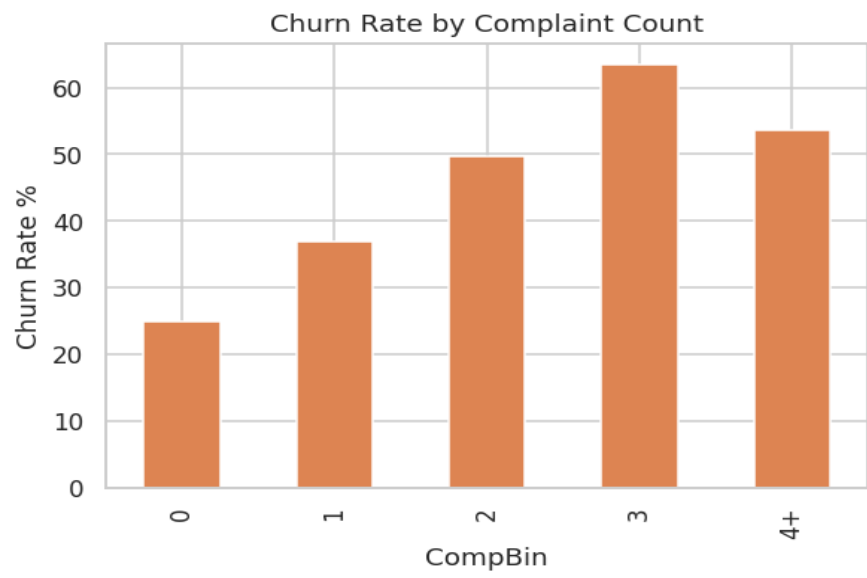
This report summarizes an end-to-end churn analysis across 7,500 customers. 31.7% of customers churned in the observed period, representing \$205,091 in monthly recurring revenue and \$8,792,436 in cumulative customer lifetime value at risk. A Logistic Regression model, selected from 8 compared algorithms, ranks customers by churn risk with a ROC-AUC of 0.7835.

## Top Churn Drivers

| Driver           | High-Risk Segment | Churn Rate | Low-Risk Segment | Churn Rate |
|------------------|-------------------|------------|------------------|------------|
| Contract Type    | Month-to-Month    | 44.3%      | Two Year         | 11.7%      |
| Internet Service | Fiber Optic       | 46.3%      | DSL              | 22.2%      |
| Complaints       | 3 complaints      | 63.4%      | 0 complaints     | 25.0%      |
| Tenure           | 0-6 months        | 51.2%      | 49+ months       | 19.5%      |
| Payment Method   | Electronic Check  | 37.2%      | Auto-pay (avg)   | 28.6%      |

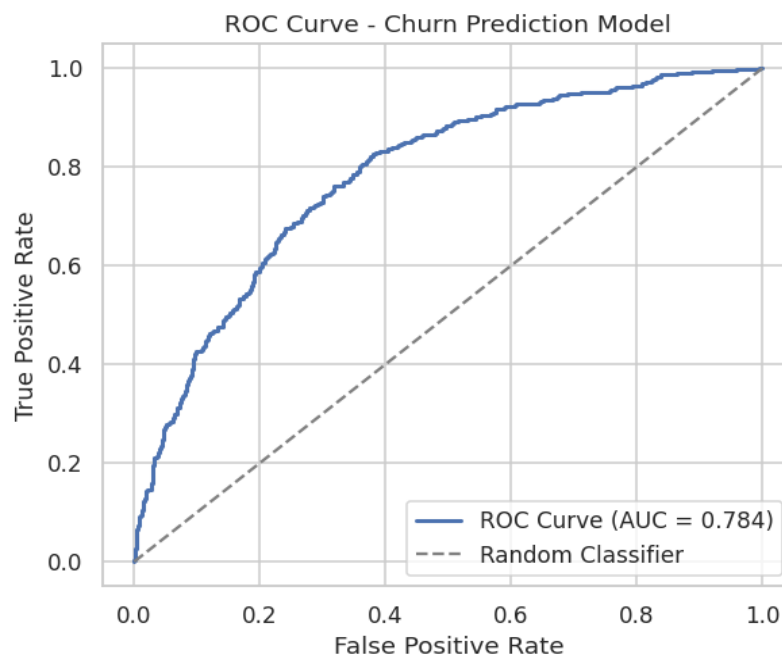
## Visual Findings

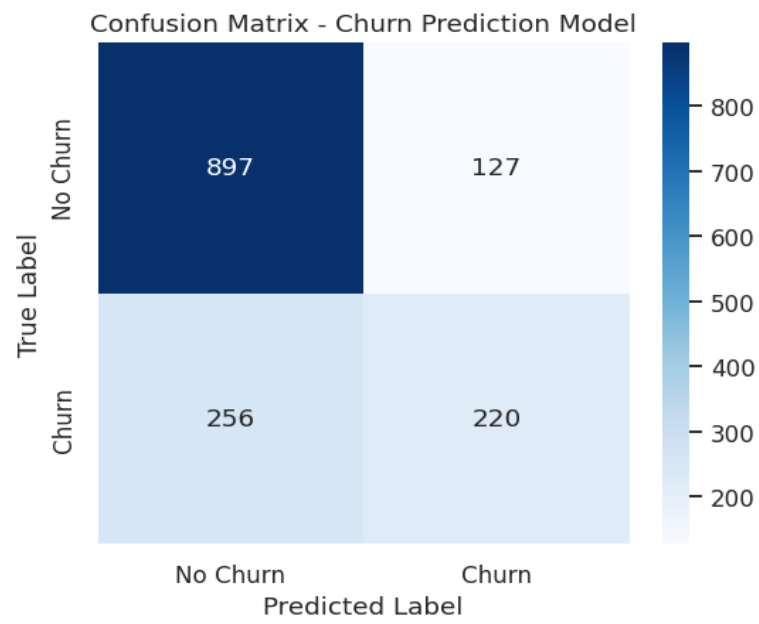




## Model Performance

| Model                        | Test ROC-AUC | CV ROC-AUC        |
|------------------------------|--------------|-------------------|
| Logistic Regression (winner) | 0.7835       | 0.7880 +/- 0.0118 |
| Gradient Boosting            | 0.7801       | 0.7840 +/- 0.0105 |
| Random Forest                | 0.7707       | 0.7771 +/- 0.0115 |
| XGBoost                      | 0.7698       | 0.7661 +/- 0.0093 |
| Naive Bayes                  | 0.7641       | 0.7700 +/- 0.0087 |
| SVM                          | 0.7550       | 0.7480 +/- 0.0057 |
| Decision Tree                | 0.7522       | 0.7494 +/- 0.0104 |
| K-Nearest Neighbors          | 0.7384       | 0.7404 +/- 0.0066 |





## Recommended Actions

1. Launch a contract-conversion incentive targeting month-to-month customers.
2. Audit the Fiber Optic customer experience (pricing, reliability, support quality).
3. Build a complaint-triggered save workflow at the 2nd complaint, before risk crosses 50%.
4. Strengthen onboarding support in the first 6 months of the customer lifecycle.
5. Incentivize Electronic Check payers to switch to autopay.
6. Bundle Tech Support / Online Security trials into new sign-ups.
7. Expand the referral program — referred customers churn 6.2 points less.

## Feature Importance

